

14. Given R3 = 0x9A23C8ABH and R4=0x42556688H, what will be the value of R3 & R4 after executing the instruction CLZ.W R3, R4?
 A) R4 = 0x9A23C8ABH & R3 = 0x9A23C8ABH B) R4 = x00000001H & R3 = 0x00000001H
 C) R4 = 0x42556688H & R3 = 0x00000001H D) R4 = x00000001H & R3 = 0x9A23C8ABH
15. Given R1 = 0x1234567BH and R0=0x5678ABCDH, what will be the value of R0 & R1 after executing the instruction UBFX.W R1,R0,#4,#8?
 A) R1= 0x000000ABH & R0=0x5678ABCDH B) R1= 0x000000CDH & R0=0x5678ABCDH
 C) R1= 0x0000008AH & R0=0x5678ABCDH D) R1= 0x000000BCH & R0=0x5678ABCDH
16. Which of the following is an operational quality attribute of an embedded system?
 A) Reliability B) Maintainability
 C) Portability D) Reusability
17. Which of the following defines a process?
 A) program in execution B) stored program
 C) function call D) thread
18. Which condition does not favor deadlock?
 A) Mutual exclusion B) Hold and wait
 C) Preemption D) Circular wait
19. The disadvantage of priority scheduling is
 A) High CPU utilization B) Starvation of lower-priority processes
 C) Low efficiency D) Low CPU utilization
20. Hardware-software co-design refers to:
 A) Designing software only B) Designing hardware only
 C) Simultaneously designing hardware and software D) Using only off-the-shelf components

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a neat diagram, explain the following: i) I ² C Bus (ii) IrDA	8	L*2	2	1
	b) Illustrate the architectural block diagram of embedded system and mention the components used.	8	L2	1	1
2.	a) With the neat diagram explain SRAM cell implementation and its working. Give comparison between SRAM and DRAM cells.	8	L2	1	1
	b) With a neat diagram, explain the architecture of ARM Cortex M3, processor.	8	L2	2	1
3.	a) With a neat diagram, explain the following: i) Watchdog Timer (ii) Brown-out Protection Circuit	8	L2	1	1
	b) Explain the Thumb-2 instruction set architecture and its benefits.	8	L2	2	1
Unit – II					
4.	a) Write an assembly language program to find the largest element from a set of 10 bytes of data stored in memory. Explain the logic used in your program.	8	L3	4	2
	b) Explain the following ARM Cortex-M3 instructions with suitable examples: SXTB, RSB.W, TST, BL	8	L2	3	1
5.	a) Write an ARM assembly program with a subroutine that computes factorial of a number stored in memory.	8	L3	4	2
	b) Explain two stack models and reset sequences in ARM Cortex M3.	8	L2	3	1

6. a) With diagram, explain the organization of CMSIS and its benefits. 8 L2 4 1
 b) Illustrate how the following ARM Cortex-M instructions modify the registers contents by showing their values before and after execution.

Address	Data value
0x00000012	0xABCDEF54H
0x0000000E	0xABCDEF54H
0x0000000A	0x24567893H
0x00000006	0x88564478H

Register	Content
R1	0x00000006
R2	0x00000000
R3	0x00000003
R4	0xFFFFFFFF

- i. LDRSB.W R2, [R1,#4]
 ii. LDRSH.W R2, [R1,#4]
 iii. SMLAL R1, R2, R3, R4
 iv. SMULL R1, R2, R3, R4

8 L3 3 2

Unit – III

7. a) Describe coin operated telephone system with a FSM, function of states and state transition diagram 8 L2 5 1
 b) Apply the Round-Robin (RR) scheduling algorithm with a 2 ms time slice to the set of processes given in the table below. Determine the Waiting Time (WT), Turnaround Time (TAT), and their averages by utilizing the algorithm. Assume no I/O operations are involved.

Process	P1	P2	P3
Arrival Time (ms)	0	0	0
Service Time(ms)	6	4	2

8 L3 5 2

8. a) Explain the operational quality attributes of embedded systems. 8 L2 5 1
 b) Apply non-preemptive priority based scheduling algorithm to the set of processes given in the table below. Determine the Waiting Time (WT), Turnaround Time (TAT), and their averages by utilizing the algorithm. Assume no I/O operations are involved.

Process	P1	P2	P3	P4
Service Time(ms)	10	5	7	6
Priority (0 = highest)	1	3	2	0
Arrival Time (ms)	0	0	0	5

8 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
